

Dharmsinh Desai University
Data Structures
Introduction to Hash Tables

Exercise:

- Create Python programs for the following.
 - Create your programs in the VS Code with complete documentation.
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1. Write a program to implement a hash table using an array data structure. Implement a suitable hash function to map keys to indices and perform basic operations such as insertion, searching, and deletion of elements. Show collision in that implementation.
 2. Extend your program to implement a hash table using the separate chaining method to handle collisions.
 3. Write a program to implement a hash table using an array data structure. Perform basic operations such as insertion, searching, and deletion of elements. Use open addressing with linear probing for collision resolution.

Practice Exercise:

- Write a program to implement a hash table using an array data structure. Perform basic operations such as insertion, searching, and deletion of elements. Use open addressing with quadratic probing for collision resolution.